

REDUCTION MODULO p AND A BOUND ON THE TORSION

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The aim is a Magma script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 1 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, section 2 turns this into two bounds, and section 3 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant Δ .
- The torsion subgroup $T := E(\mathbb{Q})_{\text{tors}}$.
- A prime p is **good** if $p \nmid \Delta$. Then reducing the equation modulo p gives an elliptic curve E_p over $\mathbb{F}_p$.
- For an abelian group G , $G[m]$ is the subgroup killed by m .
- A finite abelian group G is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leq n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

How to use this. Allow 20 minutes for §1 and 30 for §2. Section 3 is for home. Every Magma block runs in seconds in the free **online calculator**. Paste them in order, since later blocks use earlier functions.

1. REDUCTION IS INJECTIVE ON TORSION

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 1.6 is from Application VII.3.5 there.

In the following definition and results, p is a good prime for E .

Definition 1.1 (Reduction modulo p). We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \rightarrow E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

Proposition 1.2 (Minimality). *The equation is minimal at p , because $v_p(\Delta) = 0 < 12$. So E_p is the reduction of E modulo p [Sil09, Remark VII.1.1].*

Proposition 1.3 (Reduction is a homomorphism). *The reduction map $E(\mathbb{Q}_p) \rightarrow E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of O and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$ [Sil09, Proposition VII.2.1 and the discussion after it].*

Proposition 1.4 (Torsion prime to p). *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$ [Sil09, Proposition VII.3.1].*

Theorem 1.5 (Torsion of p -power order). *Let $P \in E(\mathbb{Q}_p)$ have exact order p^n with $n \geq 1$, and let $r := \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r}x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$ [Sil09, Theorem VII.3.4(b)].*

Problem 1.6 (Injectivity of reduction on torsion).

- (1) Let $p \geq 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \rightarrow E_p(\mathbb{F}_p)$$

is injective on T . In particular T is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.^a

- (2) The prime 2 is different: even if $p = 2$ is good, $T \rightarrow E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E: y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);           // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));     // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8]; 2*P eq E!0;                // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 1.4 and Theorem 1.5 fails, and what does Proposition 1.4 still give at $p = 2$?

^aHint: a nonzero torsion point in $E_1(\mathbb{Q}_p)$ has a multiple of prime order in $E_1(\mathbb{Q}_p)$.

2. TWO BOUNDS

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

Problem 2.1 (The order bound).

The first piece of the script computes the greatest common divisor b of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leq N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);           // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
E2 := EllipticCurve([0, 3]);                     // y^2 = x^3 + 3
Discriminant(E2);                               // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]]; // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                             // 1
```

- (1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.
- (2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);       // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);             // -611 -324710451
IsIsomorphic(E1, E4);                           // true
```

```
OrderBound(E1, 7), OrderBound(E4, 7); // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

Corollary 2.2 (The m -torsion). *Let E be an elliptic curve over a field K , and $m \geq 1$. If $\text{char } K \nmid m$, then $E[m](\bar{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\text{char } K = p > 0$, then $E[p^e](\bar{K})$ is cyclic for every $e \geq 1$ [Sil09, Corollary III.6.4].*

Problem 2.3 (The structure bound).

The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns $[]$ when G is trivial:

```
E3 := EllipticCurve([1, 0]); // y^2 = x^3 + x
Discriminant(E3); // -64
OrderBound(E3, 30); // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [[ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

- (1) Let G be a finite subgroup of $E(\bar{K})$, for an elliptic curve E over a field K . Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of G . Show that $\#E[n_1(G)](\bar{K}) = n_1(G)^2$, and that an integer $m \geq 1$ with $\#E[m](\bar{K}) = m^2$ satisfies $E[m](\bar{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.^a
- (2) Let b_1 and b_2 be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leq N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0; b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30); // [ 1, 2 ]
P := E3![0, 0]; 2*P eq E3!0; // true
```

Conclude that $T = \{O, (0, 0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does b stay at 4 however large N is?

- (3) Show that a rational point P is a torsion point if and only if $b_2 P = O$, and that rational torsion points generating a subgroup of order $b_1 b_2$ generate T .

^a*Hint:* for a prime ℓ , the group $G[\ell]$ lies in $E[\ell](\bar{K})$. Use Corollary 2.2.

3. HOW GOOD ARE THE BOUNDS? THE LMFDB

The [L-functions and modular forms database](#) [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

Proposition 3.1 (Reduction of isogenies). *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime p of both equations [Mil20, proof of Theorem V.4.1].*

Proposition 3.2 (Isogenous curves over $\mathbb{F}_p$). *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points [Sil09, Exercise 5.4(a)].*

Problem 3.3 (Testing the bounds).

- (1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, a -invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7s LMFDB %-9s order bound %-3s structure bound %o\n",
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1 LMFDB [ 1 ] order bound 5 structure bound [ 1, 5 ]
// 14.a2 LMFDB [ 2 ] order bound 6 structure bound [ 1, 6 ]
// 19.a2 LMFDB [ 3 ] order bound 3 structure bound [ 1, 3 ]
// 15.a4 LMFDB [ 4 ] order bound 8 structure bound [ 1, 4 ]
// 11.a2 LMFDB [ 5 ] order bound 5 structure bound [ 1, 5 ]
// 14.a6 LMFDB [ 6 ] order bound 6 structure bound [ 1, 6 ]
// 26.b2 LMFDB [ 7 ] order bound 7 structure bound [ 1, 7 ]
// 15.a8 LMFDB [ 8 ] order bound 8 structure bound [ 1, 8 ]
// 54.b2 LMFDB [ 9 ] order bound 9 structure bound [ 1, 9 ]
// 66.c3 LMFDB [ 10 ] order bound 10 structure bound [ 1, 10 ]
// 90.c7 LMFDB [ 12 ] order bound 12 structure bound [ 1, 12 ]
// 15.a2 LMFDB [ 2, 2 ] order bound 8 structure bound [ 2, 4 ]
// 15.a5 LMFDB [ 2, 4 ] order bound 8 structure bound [ 2, 4 ]
// 30.a6 LMFDB [ 2, 6 ] order bound 12 structure bound [ 2, 6 ]
// 210.e6 LMFDB [ 2, 8 ] order bound 16 structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?

- (2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]); // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]); // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
```

```
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p))))),
  Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 3.1 and 3.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.

- (3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]); P := E![5, -2]; // 15.a4
[k*P eq E!0 : k in [1..4]]; // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?

- (4) Find these curves yourself: on <https://www.lmfdb.org/EllipticCurve/Q/>, search by “torsion structure”, such as [2,4], and sort by conductor. On a curve’s page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

Remark 3.4 (Beyond the local theory). Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\text{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

REFERENCES

- [LMF25] The LMFDB Collaboration, *The L-functions and modular forms database*, 2025, <https://www.lmfdb.org>. [Online; accessed 21 July 2025]. $\uparrow$
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